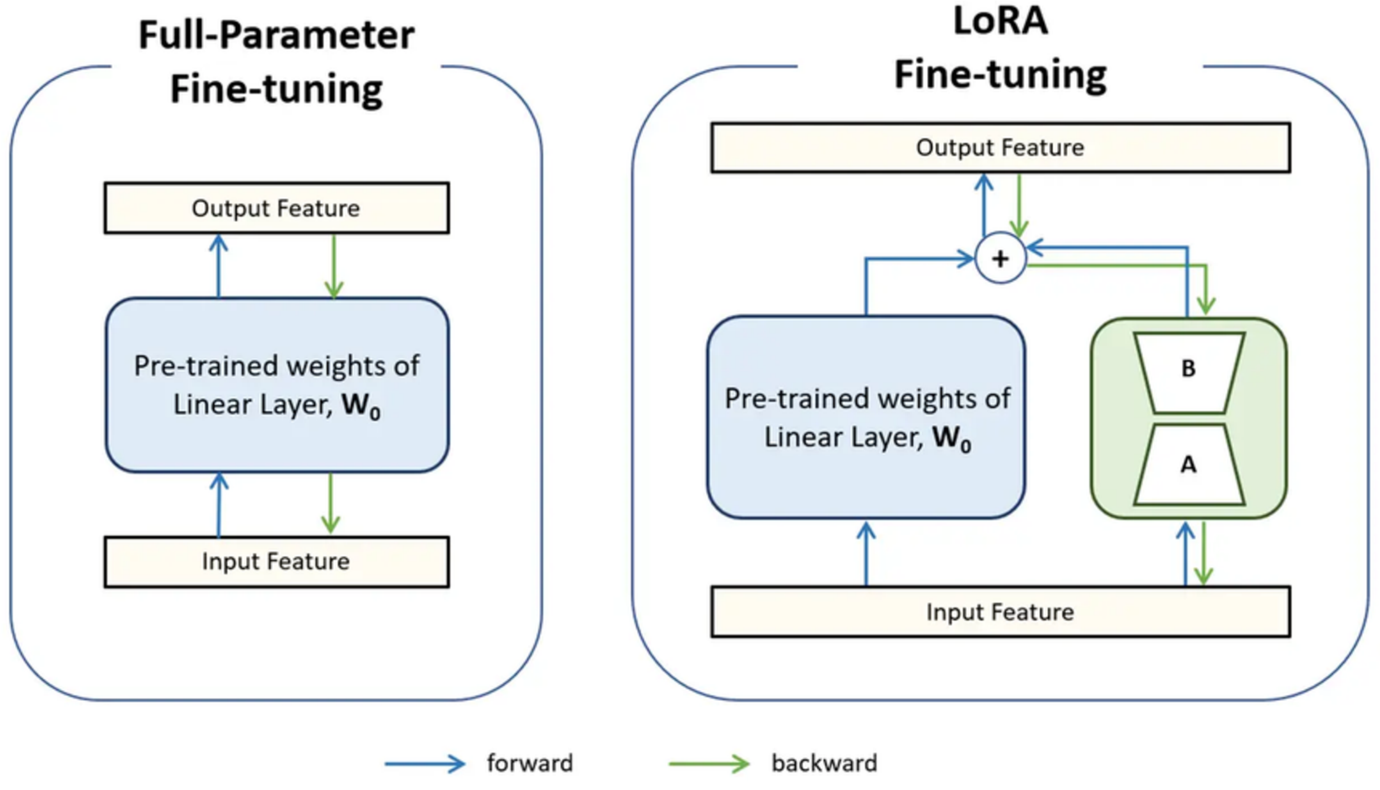
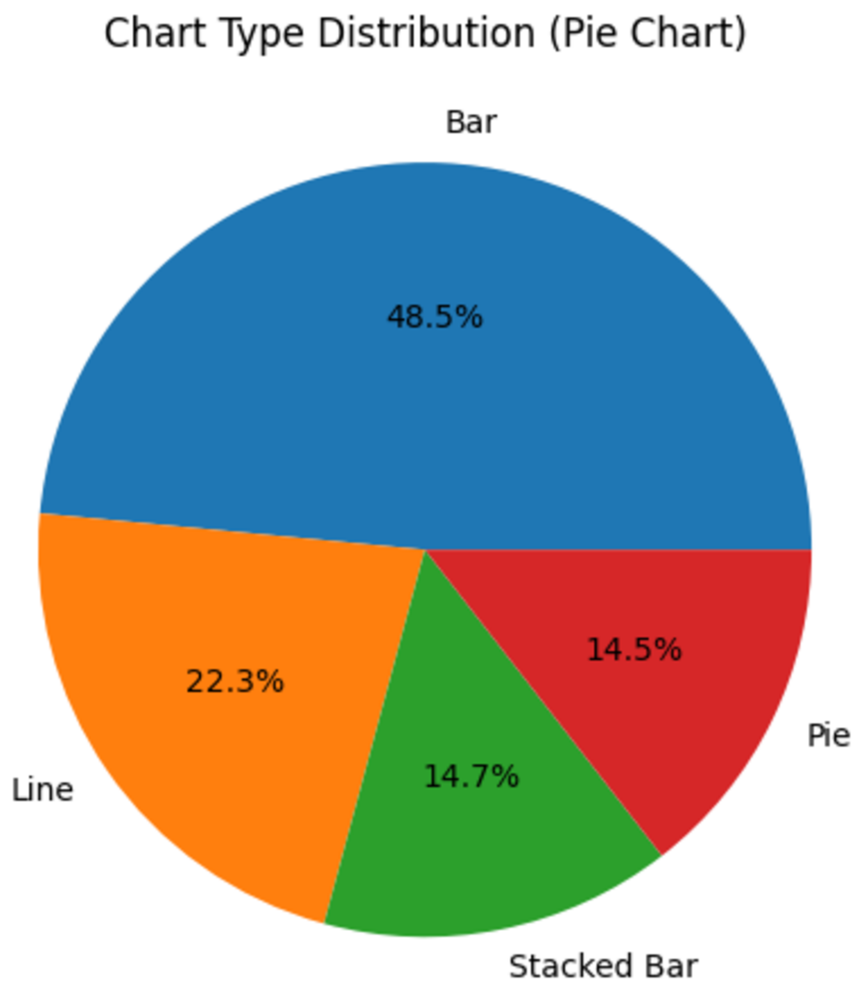
Experimental Setup
Environment



Model: Qwen2.5-VL-3B-Instruct
Platforms Used: Kaggle and Local PC
Hardware:
- NVIDIA RTX 3090 24GB VRAM (Local PC)

Training Dataset — Chart-RVR GRPO Train [15]
- Full dataset: 34,194 samples (ChartQA, PlotQA)
- Subset used: 1,000 samples - random sampling

Test Dataset — Chart-RVR GRPO Test [15]
- Total Samples Evaluated — 500
- In Distribution: ChartQA, PlotQA
- Out of Distribution: ChartFC

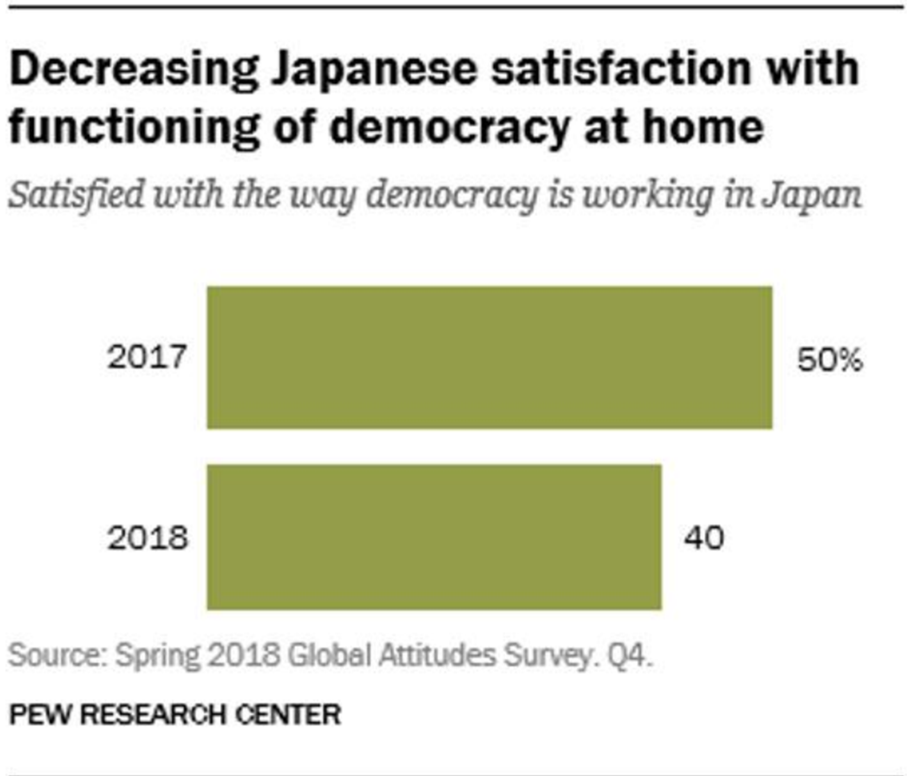


Experimental Setup

Dataset Samples

System Prompt

You are given a chart image and a query about the chart. Think step-by-step about how to answer the query based on the chart image and then provide the final answer...



Query

The difference between 2017 and 2018 in the percentage of Japanese satisfaction with democracy is what percentage?

Ground Truth : 10

Output Format

```
<type> Bar </type>

<table>
{
 "columns": ["Year", "Percentage"],
 "rows": [
 ["2017", "50"],
 ["2018", "40"] ]
}
</table>

<think>
<step-1>: The question asks for ... ...
<step-2>: From the chart: 2017 = 50%, 2018 = 40%.
<step-3>: We need to calculate the difference.
<step-4>: The difference is 50% - 40% = 10%.
<step-5>: Values taken directly from chart.
Verified.
</think>

<answer> 10 </answer>
```

Results

Performance Evaluation — Experiment List

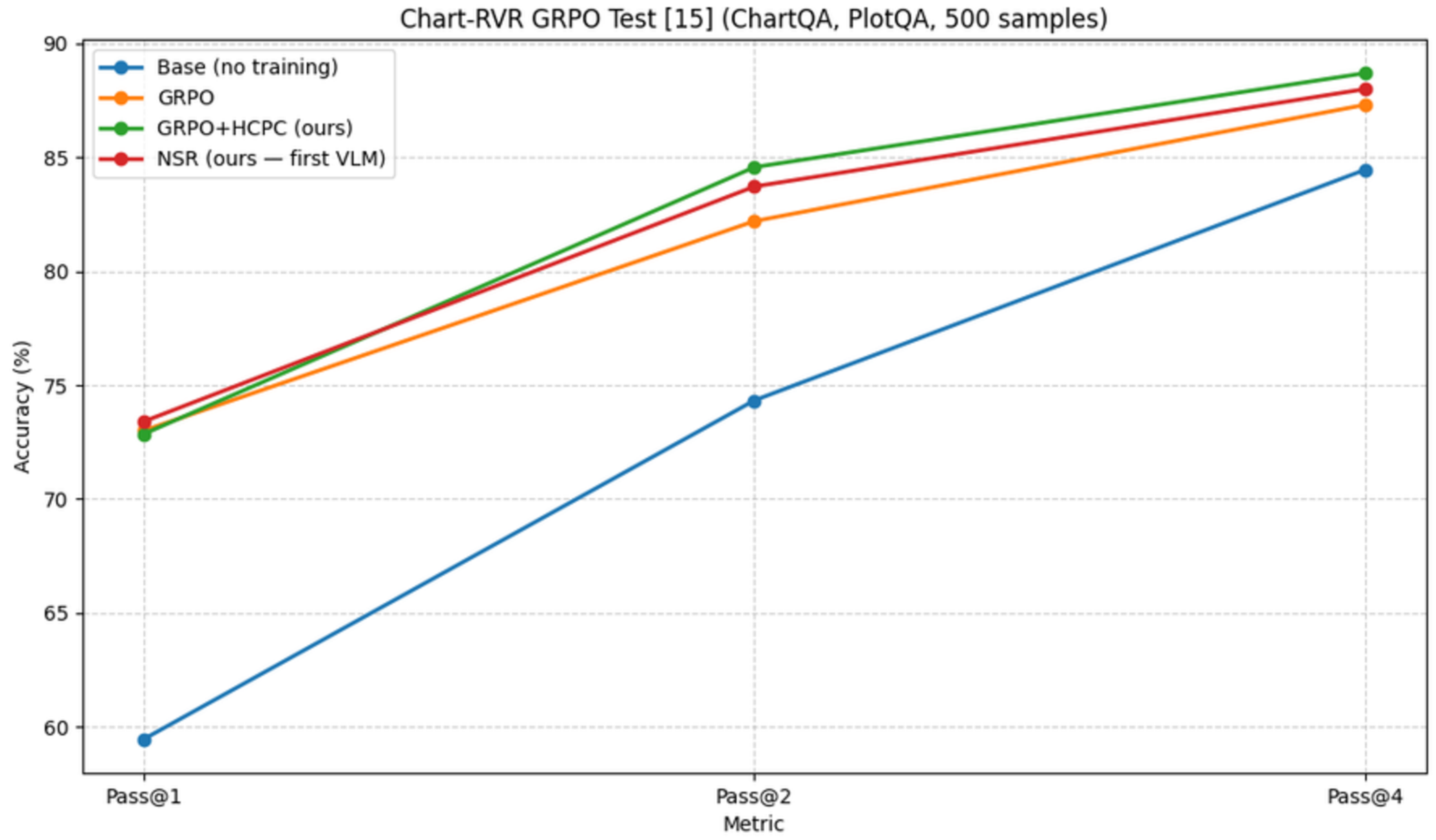
Exp. ID	Experiment	Method(s)	Purpose
E0	Base model evaluation	Qwen2.5-VL-3B-Instruct	Establish zero-training baseline
E1	Standard RLVR baseline	GRPO	Test whether flat verifiable rewards improve chart QA
E2	Proposed HCPC reward	GRPO+HCPC	Test hierarchical correct-path consistency vs flat GRPO
E3	Negative-only reinforcement baseline	NSR	Test whether text-only NSR transfers to VLM chart reasoning and compare with GRPO-HCPC
E4	OOD fact-checking transfer	GRPO, GRPO+HCPC, NSR	Test distribution shift from ChartQA-style QA to chart fact-checking

Results

Performance Evaluation

Chart-RVR GRPO Test [15] (ChartQA, PlotQA, 500 samples)

Method	Pass@1	Pass@2	Pass@4
Base (no training)	59.45	74.33	84.46
GRPO	73	82.2	87.31
GRPO+HCPC (ours)	72.85	**84.57**	**88.70**
NSR (ours — first VLM)	**73.4**	83.72	88.00

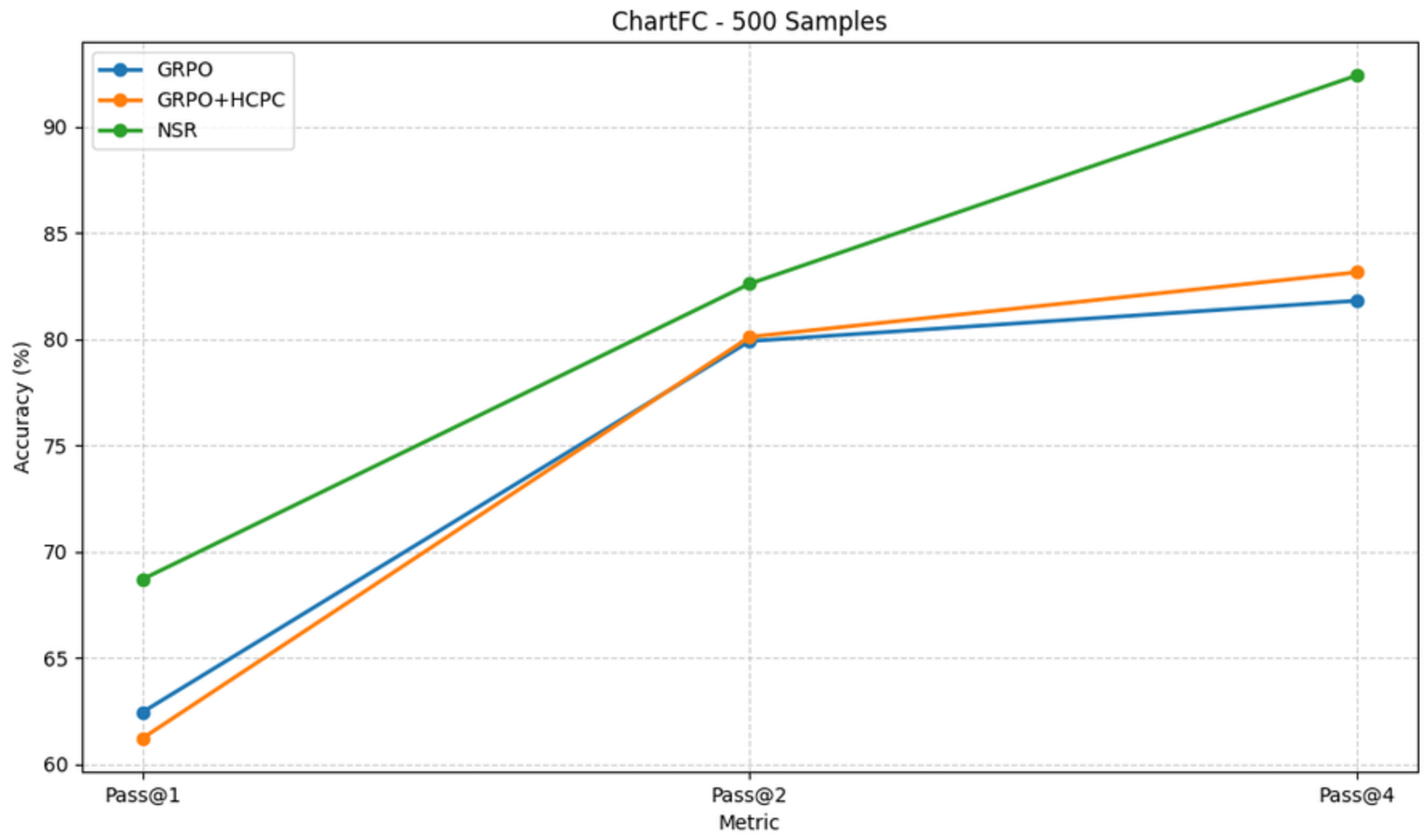


Results

Performance Evaluation

ChartFC - 500 Samples

Method	Pass@1	Pass@2	Pass@4
GRPO	62.45	79.9	81.8
GRPO+HCPC	61.22	80.1	83.14
NSR	**68.70**	**82.60**	**92.40**



Conclusion and Future Plan

What We Did

- **HCPC (Hierarchical Correct-Path Consistency):** A cross-rollout reward bonus that rewards consistent table extraction and diverse reasoning — conditioned on correctness of intermediate steps, not just final answers
- **NSR applied to VLMs for the first time:** Extends negative-only reinforcement [32] to vision-language models with noisy, multi-component rewards. Used here as a competitive baseline to test whether implicit diversity preservation (withholding gradients on correct rollouts) can match explicit diversity engineering (HCPC).

Limitations &Future Plans

- **Model scale and recency** —Extending to larger parameter regimes (7B, 72B), Conducting experiments with K = 8.
- **Full training corpus** — Current results are based on a 1,000-sample subset of the 34,194-sample Chart-RVR dataset.
- **Implement CLC and NSR-HCPC Pipelines** — Add CLC reward and NSR-HCPC reward in pipeline to enhance reasoning robustness, consistency, and alignment with desired outputs.
- **Entropy-based reasoning diversity** — A principled alternative is to directly regularize the entropy of the rollout distribution during training, enabling explicit, continuous control over diversity without dependence on surface-level textual overlap.
- **Out-of-distribution robustness** — Investigate domain-adaptive training strategies and evaluate across a broader range of chart distributions.

References

[1] Ali, A., Baheti, A., Chang, J., Chi, T.-C., Cui, B., Drozdov, A., Frankle, J., Gupta, A., Koppol, P., Kulinski, S., Li, J., Misra, D., Opsahl-Ong, K., Ortiz, J. J. G., Zaharia, M., & Zhang, Y. (2025). A state-of-the-art SQL reasoning model using RLVR. arXiv. https://arxiv.org/abs/2509.21459
[2] Ke, Z., Jiao, F., Ming, Y., Nguyen, X.-P., Xu, A., Long, D. X., Li, M., Qin, C., Wang, P., Savarese, S., Xiong, C., & Joty, S. (2025). A survey of frontiers in LLM reasoning: Inference scaling, learning to reason, and agentic systems. Transactions on Machine Learning Research. https://arxiv.org/abs/2504.09037
[3] Masry, A., Puri, A., Hashemi, M., Rodriguez, J. A., Thakkar, M., Mahajan, K., Yadav, V., Madhusudhan, S. T., Piché, A., Bahdanau, D., Pal, C., Vazquez, D., Hoque, E., Taslakian, P., Rajeswar, S., & Gella, S. (2025). BIGCHARTS-R1: Enhanced chart reasoning with visual reinforcement finetuning. *COLM*. https://arxiv.org/abs/2508.09804
[4] Odena, A., Lawson, D., & Olah, C. (2017). Changing model behavior at test-time using reinforcement learning. *ICLR Workshop*. https://arxiv.org/abs/1702.07780
[5] Sinha, S., Frunza, O., Rasul, K., Nevmyvaka, Y., & Zhang, A. (2025). Chart-RVR: Reinforcement learning with verifiable rewards for explainable chart reasoning. arXiv. https://arxiv.org/abs/2510.10973
[6] Chen, L., Zhao, X., Zeng, Z., Huang, J., Zhong, Y., & Ma, L. (2025). Chart-R1: Chain-of-thought supervision and reinforcement for advanced chart reasoner. arXiv. https://arxiv.org/abs/2507.15509
[7] Carbune, V., Mansoor, H., Liu, F., Aralikatte, R., Baechler, G., Chen, J., & Sharma, A. (2024). Chart-based reasoning: Transferring capabilities from LLMs to VLMs. *NAACL Findings.* https://aclanthology.org/2024.findings-naacl.62/
[8] Kaur, R., Srishankar, N., Zeng, Z., Ganesh, S., & Veloso, M. (2025). ChartAgent: A multimodal agent for visually grounded reasoning in complex chart question answering. arXiv. https://arxiv.org/abs/2510.04514
[9] Goswami, K., Mathur, P., Rossi, R., & Dernoncourt, F. (2025). ChartCitor: Answer citations for ChartQA via multi-agent LLM retrieval. *ACM Web Conference.* https://doi.org/10.1145/3701716.3716886
[10] Masry, A., Thakkar, M., Bajaj, A., Kartha, A., Hoque, E., & Joty, S. (2025). ChartGemma: Visual instruction-tuning for chart reasoning in the wild. *COLING Industry Track*. https://arxiv.org/abs/2407.04172
[11] Masry, A., Shahmohammadi, M., Parvez, M. R., Hoque, E., & Joty, S. (2024). ChartInstruct: Instruction tuning for chart comprehension and reasoning. *ACL Findings.* https://aclanthology.org/2024.findings-acl.619/
[12] Xu, Z., Du, S., Qi, Y., Lu, S., Xu, C., Yuan, C., & Guo, J. (2025). ChartPoint: Guiding MLLMs with grounding reflection for chart reasoning. *ICCV*. https://arxiv.org/abs/2512.00305
[13] Hegde, S., Fazli, P., & Seifi, H. (2026). ChartQA-X: Generating explanations for visual chart reasoning. *WACV*. https://arxiv.org/abs/2504.13275
[14] Masry, A., Long, D. X., Tan, J. Q., Joty, S., & Hoque, E. (2022). ChartQA: A benchmark for question answering about charts with visual and logical reasoning. *ACL Findings*. https://aclanthology.org/2022.findings-acl.177/

References

[15] Xia, R., Zhang, B., Ye, H., Yan, X., Liu, Q., Zhou, H., Chen, Z., Dou, M., Shi, B., Yan, J., & Qiao, Y. (2024). ChartX and ChartVLM: A versatile benchmark and foundation model for complicated chart reasoning. *IEEE Transactions on Image Processing*. https://arxiv.org/abs/2402.12185

[16] Su, Y., Yu, D., Song, L., Li, J., Mi, H., Tu, Z., Zhang, M., & Yu, D. (2025). Crossing the reward bridge: Expanding RL with verifiable rewards across diverse domains. arXiv. https://arxiv.org/abs/2503.23829

[17] Zhang, Y., Yang, J., Yuan, Y., & Yao, A. C.-C. (2024). Cumulative reasoning with large language models. TMLR. https://arxiv.org/abs/2308.04371

[18] He, W., Xi, Z., Zhao, W., Fan, X., Ding, Y., Shan, Z., Gui, T., Zhang, Q., & Huang, X. (2025). Distill visual chart reasoning ability from LLMs to MLLMs. *EMNLP Findings.* https://aclanthology.org/2025.findings-emnlp.172/

[19] Liu, J., He, C., Lin, Y., Yang, M., Shen, F., Liu, S., & Gao, T. (2025). ETTRL: Balancing exploration and exploitation in LLM test-time RL via entropy mechanism. arXiv. https://arxiv.org/abs/2508.11356

[20] Su, Y., et al. (2025). Expanding RL with verifiable rewards across diverse domains. arXiv. https://arxiv.org/abs/2503.23829

[21] Köksal, A., & Alatan, A. A. (2025). Few-shot vision-language reasoning for satellite imagery via verifiable rewards. *ICCV.* https://arxiv.org/abs/2507.21745

[22] Teney, D., Liu, L., & van den Hengel, A. (2017). Graph-structured representations for visual question answering. *CVPR*. https://arxiv.org/abs/1609.05600

[23] Duan, W., Lu, J., & Xuan, J. (2024). Group-aware coordination graph for multi-agent reinforcement learning. *IJCAI*. https://doi.org/10.24963/ijcai.2024/434

[24] Zhang, R., et al. (2025). Improve vision-language model chain-of-thought reasoning. *ACL*. https://aclanthology.org/2025.acl-long.82/

[25] Lu, P., Gong, R., Jiang, S., Qiu, L., Huang, S., Liang, X., & Zhu, S.-C. (2021). Inter-GPS: Interpretable geometry problem solving with formal language and symbolic reasoning. *ACL-IJCNLP.* https://aclanthology.org/2021.acl-long.528/

[26] Wen, X., et al. (2025). Reinforcement learning with verifiable rewards implicitly incentivizes correct reasoning in base LLMs. arXiv. https://arxiv.org/abs/2506.14245

[27] Zhang, Z., et al. (2025). RLVMR: Reinforcement learning with verifiable meta-reasoning rewards. arXiv. https://arxiv.org/abs/2507.22844

[28] Akhtar, M., Cocarascu, O., & Simperl, E. (2023). Reading and reasoning over chart images for evidence-based fact-checking. *EACL Findings*. https://aclanthology.org/2023.findings-eacl.30/

[29] Zhang, T., et al. (2023). TEMPERA: Test-time prompt editing via reinforcement learning. *ICLR*. https://arxiv.org/abs/2211.11890

[30] Zuo, Y., et al. (2025). TTRL: Test-time reinforcement learning. arXiv. https://arxiv.org/abs/2504.16084

[31] Du, Y., et al. (2025). Test-time reinforcement learning for GUI grounding via region consistency. arXiv. https://arxiv.org/abs/2508.05615

[32] Zhu, X., et al. (2025). The surprising effectiveness of negative reinforcement in LLM reasoning. *NeurIPS*. https://arxiv.org/abs/2506.01347

[33] Huang, M., et al. (2025). VProChart: Answering chart question through visual perception alignment agent and programmatic reasoning. *AAAI*. https://doi.org/10.1609/aaai.v39i4.32384

[34] Yu, X., et al. (2025). Visual document understanding and reasoning: A multi-agent collaboration framework with adaptive test-time scaling. arXiv. https://arxiv.org/abs/2508.03404